

Project 1 Report

The proposed project uses a linked list. Could you explain why linked list suggested?

- I think the linked list was suggested primarily for us to understand one of the fundamental data structures. Since this was one of the first data structures introduced in class, it was important for us to practice implementing and using a linked list in a real-world project. A linked list also works well for this project because mentorship sessions can be added and removed as needed. Instead of having to shift other elements when adding or removing a session, we can change the connecting nodes

What will be another alternative solution?

- I think another solution for this project is using an array instead of a linked list. Using an array would allow us to easily access sessions using an index and would be relatively simple to implement. However, adding or removing sessions from the middle of an array can require the other elements to be shifted. Because of this, a linked list can be more convenient for a project where sessions may frequently be added or removed.

2. Explain your design:

a. How did you design your linked list.

- I took inspiration from the Project 1 supplement and designed my linked list around the requirements given for the project. My linked list contains a head node that keeps track of the first session in the list. Each node contains a Session object and a reference to the next node. The MyLinkedList class contains the required methods for adding, searching, removing, displaying, and registering participants.
- For adding sessions, I implemented addFirst, addLast, and insertAfter. For searching, I implemented searchByID and searchByMentor. I also implemented remove, registerParticipant, and display to handle the other required functionality of the program.
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b. How many classes did you use?

- I took inspiration from the Project 1 supplement and made the classes that were a requirement along with the required method for each class. I did not create any new classes other than the ones provided by the supplement.

c. Did you add any helper functionalities

- I did not use any other helper functionalities

i. Explain them if you did